The value of a term commutes with substitution

The Open Logic Project contributors

openlogic:first-order-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment prop:ext-terms (source lines 920-924, the \ollabel on line 920); the proof environment begins two lines below it and is its own proof, closing at source line 959.

Editorial note. The proof's composite case is displayed inside a multline with an aligned block; both environments are part of the retained source range and are reproduced as drawn.

Editorial note. Neither the statement nor the proof contains an \olref, an \Olref or a \cref, so no source pointer and no bibliography entry are needed; the value, substitution and assignment notations are all declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's first-order syntax and semantics unit, counted once: term evaluation commutes with substitution.

Allowed standard prerequisites (editorial).

- Term and its value under an assignment. Terms are built from constants, variables and function symbols; the value of a term in a structure M under an assignment s is defined by the obvious recursion on the term.. The induction follows the term construction and the value function is unfolded twice in the composite case (source lines 926-955).
- Substitution of a term for a variable. The result of substituting the term t' for the variable x throughout the term t, defined by recursion on t, leaving constants in place and replacing exactly the variable x.. The substitution is computed in each of the four cases (source lines 929, 933, 938, 946).
- Modified variable assignment. The assignment that agrees with s everywhere except at x, where it takes a prescribed value.. The right-hand side of the identity uses the modified assignment at every case, and the

variable cases are decided by which variables the two assignments agree on (source lines 934-940).

Proposition 0.1. *Let* $\mathfrak{A}M$ *be !!astructure,* t *and* t' *terms, and* s *a variable assignment. Then* $\mathrm{val}(t[t'/x])M[s] = \mathrm{val}(t)M[s[\mathrm{val}(t')M[s]/x]]$.

Proof. By induction on t.

1. If t is a constant, say, $t \,\mathrm{id}\, c$, then $t[t'/x] = c$, and $\mathrm{val}(c)M[s] = scM = \mathrm{val}(c)M[s[\mathrm{val}(t')M[s]/x]]$.
2. If t is a variable other than x, say, $t \,\mathrm{id}\, y$, then $t[t'/x] = y$, and $\mathrm{val}(y)M[s] = \mathrm{val}(y)M[s[\mathrm{val}(t')M[s]/x]]$ since $\sigma ss[\mathrm{val}(t')M[s]/x]x$.
3. If $t \,\mathrm{id}\, x$, then $t[t'/x] = t'$. But $\mathrm{val}(x)M[s[\mathrm{val}(t')M[s]/x]] = \mathrm{val}(t')M[s]$ by definition of $s[\mathrm{val}(t')M[s]/x]$.
4. If $t \,\mathrm{id}\, \mathrm{Atom}ft_1, \ldots, t_n$ then we have:

$$
\begin{aligned}
&\mathrm{val}(t[t'/x])M[s] = \\
&= \mathrm{val}(\mathrm{Atom}ft_1[t'/x], \ldots, t_n[t'/x])M[s] \\
&\qquad \text{by definition of } t[t'/x] \\
&= sfM(\mathrm{val}(t_1[t'/x])M[s], \ldots, \mathrm{val}(t_n[t'/x])M[s]) \\
&\qquad \text{by definition of } \mathrm{val}(\mathrm{Atom}f\ldots)M[s] \\
&= sfM(\mathrm{val}(t_1)M[s[\mathrm{val}(t')M[s]/x]], \ldots, \mathrm{val}(t_n)M[s[\mathrm{val}(t')M[s]/x]]) \\
&\qquad \text{by induction hypothesis} \\
&= \mathrm{val}(t)M[s[\mathrm{val}(t')M[s]/x]] \text{ by definition of } \mathrm{val}(\mathrm{Atom}f\ldots)M[s[\mathrm{val}(t')M[s]/x]]
\end{aligned}
$$

□